SAL EDUCATION

SAL COLLEGE OF ENGINEERING

SYLLABUS FOR MID SEM EXAM – SUMMER-2026

SUBJECT: DISCRETE MATHEMATICS AND GRAPH THEORY

SUBJECT CODE: BE04000261

SEMESTER: 4 BRANCH: I.T,COMPUTER AND CSE(DS)

MODULE	TOPIC	COURSE OUTCOME
2	**Set Theory**: Basic concepts of set theory like inclusion, complement, intersection, union, Cartesian product, power set. **Counting and Combinatorics:** Basic counting principles, permutations and combinations, binomial coefficients and the binomial theorem, Pigeonhole Principle, inclusion-exclusion principle.	**CO2**
3	**Functions:** Basic concepts of functions like domain, range, subjective, injective, bijective, and composition of functions, inverse of functions.	**CO3**
4	**Relations:** Definition, binary relation, representation, domain, range, universal relation, void relation, union, intersection, and complement operations on relations, properties of binary relations in a set: reflexive, symmetric, transitive, anti-symmetric relations, partition and covering of a set, equivalence relation, equivalence classes, compatibility relation, maximum compatibility block, composite relation, converse of a relation, transitive closure of a relation R in set X **Partial Ordering:** Definition, examples, simple or linear ordering, totally ordered set (chain), frequently used partially ordered relations, representation of partially ordered sets, Hesse diagrams, least & greatest elements, minimal & maximal elements, least upper bound (Supremum),	**CO4**

	greatest lower bound (Infimum), Well-ordered partially ordered sets (Posets). Lattice as Posets, complete, distributive, modular and complemented lattices, Boolean and pseudo Boolean lattices (definitions and simple examples only).	
5	**Graphs:** Introduction, definition, examples; Nodes, edges, adjacent nodes, directed and undirected edge, directed graph, undirected graph, examples; Initiating and terminating nodes, loop (sling), distinct edges, parallel edges, multi-graph, simple graph, weighted graphs, examples; Isolated nodes, null graph, isomorphic graphs, examples; Degree, In-degree, out-degree, total degree of a node, examples; Subgraphs: definition, examples; Converse(reversal or directional dual) of a digraph, examples; Path: Definition, paths of a given graph, length of path, examples; Simple path (edge simple), elementary path (node simple), examples; Cycle (circuit), elementary cycle, examples; **Reachability:** Definition, geodesic, distance, examples; Properties of reachability, the triangle inequality; Reachable set of a given node, examples; Node base, examples	**CO5**

SUBJECT COORDINATOR

DR. SWETA BHATT